

## Savitzky-Golay optimizer diagnostics

Objective: mean squared translational/angular acceleration residual; angular residual uses the reference inertia/mass metric

Tolerances: {'ftol': None, 'xtol': None, 'gtol': 1e-06, 'note': 'ftol and xtol are disabled by default in the SG estimator; gtol or max\_nfev terminates the solve unless explicitly overridden'}

Selected strict-ZOH solve

status=0 success=False nfev=80 njev=6

message=The maximum number of function evaluations is exceeded.

reported optimality=7.88752095104428e-06

initial={'cost': 583.2421632430004, 'gradient\_inf\_norm': 0.01569152974195731, 'gradient\_l2\_norm': 0.017541869922718285, 'jacobian': {'condition\_number\_nonzero\_subspace': 324.5677178162147, 'numerical\_rank': 14, 'numerical\_tolerance': 1.6698122317628218e-10, 'singular\_values': [191.54777750243863, 174.73326607317125, 152.6688903509138, 48.52381283874322, 44.760278213303486, 25.941479744077085, 13.174578223969108, 11.997446999567352, 11.665379483169488, 10.772322107016365, 9.272683530978469, 3.817025075898144, 3.7877080783348736, 0.5901627518325833]}, 'residual\_l2\_norm': 34.15383326196345}

final={'cost': 583.2421630275993, 'gradient\_inf\_norm': 7.88752095104428e-06, 'gradient\_l2\_norm': 1.1155482309803043e-05, 'jacobian': {'condition\_number\_nonzero\_subspace': 324.730746272064, 'numerical\_rank': 14, 'numerical\_tolerance': 1.6698152844829302e-10, 'singular\_values': [191.54812768657368, 174.7333336131906, 152.6689203078987, 48.52384630173439, 44.76035804439255, 25.941533211511825, 13.174555771813676, 11.997415402037637, 11.665373759925682, 10.772338384570281, 9.272815461239842, 3.8170567145487704, 3.7876868155933456, 0.5898675437591363]}, 'residual\_l2\_norm': 34.15383325565666}

coordinate\_step\_l2\_norm=0.00028055998027753763

cost\_reduction=2.1540108718909323e-07

gradient\_tolerance\_satisfied=False

finite\_difference\_max\_relative\_error=5.1508985782697236e-12

finite\_difference\_directional\_checks=[{'analytic\_l2\_norm': 177.87289742727805, 'difference\_l2\_norm': 9.1620525447151e-10, 'direction': 'negative\_gradient', 'finite\_difference\_l2\_norm': 177.8728974273998, 'relative\_error': 5.1508985782697236e-12, 'scheme': 'central', 'step': 6.0554544523933395e-06}]

All smoothstep and strict-ZOH stage details are in optimizer\_diagnostics.json.